

Chapter 5

Steady Electric Currents



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Static electric field problems

Starting from

Maxwell equations

$$\begin{cases} \nabla \times \vec{E} = 0 \\ \nabla \cdot \vec{D} = \rho \end{cases}$$

$$\nabla \cdot \vec{J} = 0$$

Conditions

Constitutive relations

$$\begin{cases} \vec{D} = \varepsilon \vec{E} \\ \vec{J} = \sigma \vec{E} \end{cases}$$

boundary conditions

$$\begin{cases} \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \end{cases}$$

$$\vec{e}_n \cdot (\vec{J}_1 - \vec{J}_2) = 0$$

Chap 5

Classification according to the charge static or motion

Electrostatic
field

Static

σ Arbitrarily

$$\vec{J} = 0$$

Uniform motion

σ Finite

$$\vec{J} \neq 0$$

Steady current
field



Contents

- 5.1 Introduction**
- 5.2 Current Density and Ohm's Law**
- 5.3 Electromotive Force and KVL**
- 5.4 Equation of Continuity and KCL**
- 5.5 Power Dissipation and Joule's Law**
- 5.6 Boundary Conditions for Current Density**
- 5.7 Resistance Calculations**

5.1 Introduction

- There are several types of electric currents caused by the motion of free charges
- **Electrolytic currents (电解电流):** migration of positive and negative ions.
- **Convection currents (运流电流):** results from motion of positively or negatively charged particles in a vacuum or rarefied gas.
- **Conduction currents (传导电流) in conductors and semiconductor:** drift motion of conduction electrons and/or holes.



5.2 Current Density and Ohm's Law

The amount of charge flowing across a given area per unit time is called the **electric current intensity** or **electric current**, and it is denoted by I . The unit of electric current is **ampere (A)**, and its density is J in Amperes per square meter. It is defined as,

$$\Delta Q = Nqu \cdot \mathbf{a}_n \Delta s \Delta t$$

$$\mathbf{J} = Nqu$$

Where N is the number of charge per unit volume, $\mathbf{u}$ is the velocity of charges, Δs is where the charges passing through.

Current models in different distributions

Current models

1. Volumn current

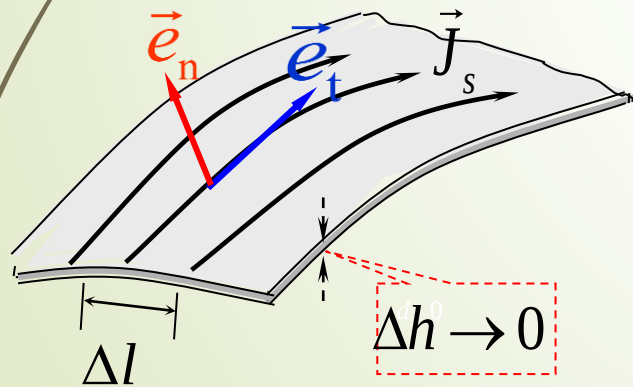
$$\vec{J} = \vec{e}_i \lim_{\Delta S \rightarrow 0} \frac{\Delta i}{\Delta S_i} = \vec{e}_i \frac{di}{dS_i}$$

movement direction
of positive charge

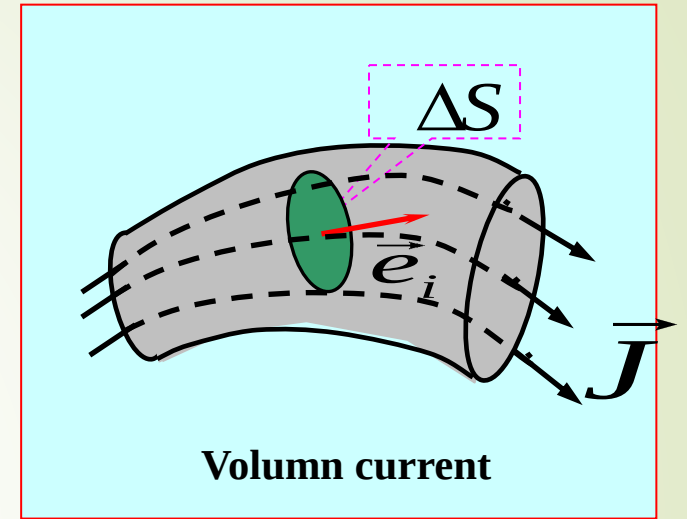
unit cross
section area

Unit: A/m² (安/米²)

The corresponding total current is: $I = \int_S \vec{J} \cdot d\vec{S}$



Surface current



Volumn current

2. Surface current

$$|\vec{J}_s| = \lim_{\Delta l \rightarrow 0} \frac{\Delta I}{\Delta l} = \frac{dI}{dl} \quad \text{Unit: A/m}$$

discussion:

$$\Delta I = \vec{J} \cdot \Delta \vec{S} = \vec{J} \cdot \hat{n} h \Delta l = (\vec{J} h) \cdot \hat{n} \Delta l = \vec{J}_s \cdot \hat{n} \Delta l$$

$$\vec{J}_s = \lim_{\substack{h \rightarrow 0 \\ J \rightarrow \infty}} h \vec{J} \neq 0$$

3. line current

$$\vec{J} = \vec{e}_i I \delta(r - r')$$

Ohm's Law

Ohm's law

$$U = IR$$



Integral form

In **most** conducting media, the **conduction** current density J at a point is **proportional** to the electric field intensity E at that point so that

$$J = \sigma E$$



Differential form
of Ohm's law

σ : **conductivity**
conducting ability of the medium

The above equations are valid **only** for a **linear isotropic** medium.

About σ

σ

$\sigma \rightarrow \infty$ P.E.C. (perfect electric conductor)

- ❖ A conductor with infinite σ is called a
- ❖ In a perfect electric conductor, electric current can be produced **without** the influence of an electric field.
- ❖ There is **no** steady electric field in a perfect electric conductor. Otherwise, an infinite current will be generated, and it results in an infinite energy.

$\sigma \rightarrow 0$ P.E.D. (perfect electric dielectric)

- ❖ A medium without any conductivity is called a perfect dielectric or an insulator.
- ❖ In nature there exists no any p.e.c. or perfect dielectric.

conductivity

The conductivities of several media unit in S/m

Media	Conductivities	Media	Conductivities
Silver	6.17×10^7	Sea water	4
Copper	5.80×10^7	Pure water	10^{-3}
Gold	4.10×10^7	Dry soil	10^{-5}
Aluminum	3.54×10^7	Transformer oil	10^{-11}
Brass	1.57×10^7	Glass	10^{-12}
Iron	10^7	Rubber	10^{-15}

Attention: convection current does not obey Ohm's Law

The magnitude of the current density of the convection current is not proportional to the electric field intensity, and the direction may be different from that of electric field intensity.

If the charge density is ρ , and the moving velocity is v , and then

$$\mathbf{J} = \rho \mathbf{v}$$



Characteristics of Steady current field

Q1: what are the physical quantities?

Q2: what characteristics of the quantities?

5.4 Equation of Continuity and KCL

- The principle of **conservation of charge** is one of the **fundamental postulates** of physics. Electric charges may not be created or destroyed;
- ❖ Consider an arbitrary volume V bounded by surface S . A net charge Q exists within this region. If a net current I flows across the surface out of this region, the charge in the volume must **decrease at a rate that equals** the current;

$$I = -\frac{dq}{dt} = -\frac{d}{dt} \int_V \rho dV$$

- ❖ Also, the current leaving the region is the total outward flux of the current density vector through the surface S ;

$$I = \oint_S \mathbf{J} \cdot d\mathbf{S}$$

Assume the density of the sustained charges in the volume V bound by the closed surface S is ρ , then

$$q = \int_V \rho dV$$

then

$$I = \oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{\partial q}{\partial t} = -\int_V \frac{\partial \rho}{\partial t} dV$$

By using the divergence theorem, we obtain

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

which is called the **principle of current continuity** in differential form.

Because the distribution of the charges in the steady electric current field is **independent of time**, i.e. $\frac{\partial \rho}{\partial t} = 0$, we find

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = 0 \quad \text{Or KCL:} \quad \sum_j I_j = 0$$

which states that in the steady electric current field the **flux** of the current density through any closed surface will be **zero**.

With differential form,

$$\nabla \cdot \mathbf{J} = 0$$

which states that the steady electric current field is **solenoidal**.



The **steady electric field** caused by the sustained charges on the plates is **also a conservative field**, and the line integral of it around a closed circuit should be zero, i.e.

$$\oint_l \mathbf{E} \cdot d\mathbf{l} = 0$$

In the conducting medium, $\mathbf{J} = \sigma \mathbf{E} \longrightarrow \oint_l \frac{\mathbf{J}}{\sigma} \cdot d\mathbf{l} = 0$

For **homogeneous media**, the above equation becomes

$$\oint_l \mathbf{J} \cdot d\mathbf{l} = 0 \quad \text{with differential form:} \quad \nabla \times \mathbf{J} = 0$$

In **homogeneous** conducting media the steady electric current field is **irrotational**.

5.5 Power Dissipation and Joule's Law

► The work Δw done by an electric field $\mathbf{E}$ in moving a charge q a distance $\Delta \vec{\ell}$ is $q\mathbf{E} \cdot (\Delta \vec{\ell})$, which corresponds to a power

$$p = \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} = q\mathbf{E} \cdot \mathbf{u},$$

where $\mathbf{u}$ is the drift velocity. The total power delivered to all the charge carriers in a volume element dv is

$$dP = \sum_i p_i = \mathbf{E} \cdot \left(\sum_i N_i q_i \mathbf{u}_i \right) dv,$$

$$dP = \mathbf{E} \cdot \mathbf{J} dv$$

$$\frac{dP}{dv} = \mathbf{E} \cdot \mathbf{J} \quad (\text{W/m}^3)$$

Power dissipation
density

$$p = \mathbf{E} \cdot \mathbf{J}$$



❖ For a given volume V , the total electric power converted into heat is

$$P = \int_V \mathbf{E} \cdot \mathbf{J} dV \quad (W)$$

➤ This is known as **Joule's law**.

❖ In a conductor of a constant cross section,

$$dv = ds d\ell$$

❖ with $d\ell$ measured in the direction $\mathbf{J}$.

$$P = \int_L E d\ell \int_S J ds = VI,$$

❖ where I is the current in the conductor. Since $V = RI$, we have

$$P = I^2 R \quad (W)$$



5.6 Boundary Condition for Current Density

1. Basic equations (general problems)

$$\begin{array}{ll} \text{Differential form :} & \begin{cases} \nabla \cdot \vec{J} = 0 \\ \nabla \times \vec{E} = 0 \\ (\nabla \cdot \vec{D} = \rho) \end{cases} & \text{Integral forms :} & \begin{cases} \oint_S \vec{J} \cdot d\vec{S} = 0 \\ \oint_C \vec{E} \cdot d\vec{l} = 0 \\ (\oint_S \vec{D} \cdot d\vec{S} = q) \end{cases} \end{array}$$

$$\text{Constitutive relations: } \vec{J} = \sigma \vec{E} \quad (\vec{D} = \epsilon \vec{E})$$

2. Boundary conditions (general problems)

$$\begin{array}{ll} \begin{cases} \vec{e}_n \cdot (\vec{J}_1 - \vec{J}_2) = 0 \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \end{cases} & \text{OR} & \begin{cases} J_{1n} - J_{2n} = 0 \\ E_{1t} - E_{2t} = 0 \\ D_{1n} - D_{2n} = \rho_s \end{cases} \end{array}$$

3. Two types of problems according to the medium (special problems)

➤ Conductive
medium :

$$\sigma \neq 0$$

➤ Existing medium : $\sigma = 0$

■ φ of steady current field

- The potential function satisfies the requirement of Laplace's equations

$$\nabla^2 \varphi = 0 \quad (\vec{E} = -\nabla \varphi)$$

- Boundary conditions of electric potential

$$\begin{cases} \varphi_1 = \varphi_2 \\ \sigma_1 \frac{\partial \varphi_1}{\partial n} = \sigma_2 \frac{\partial \varphi_2}{\partial n} \end{cases} \quad \leftarrow \quad \begin{cases} E_{1t} - E_{2t} = 0 \\ J_{1n} - J_{2n} = 0 \end{cases}$$

$$\left(\varepsilon_1 \frac{\partial \varphi_1}{\partial n} - \varepsilon_2 \frac{\partial \varphi_2}{\partial n} = -\rho_s \right) \quad (D_{1n} - D_{2n} = \rho_s)$$



■ two conductors

- Boundary conditions on both sides of the boundary

$$\begin{cases} J_{1n} - J_{2n} = 0 \\ E_{1t} - E_{2t} = 0 \quad (D_{1n} - D_{2n} = \rho_s) \end{cases}$$

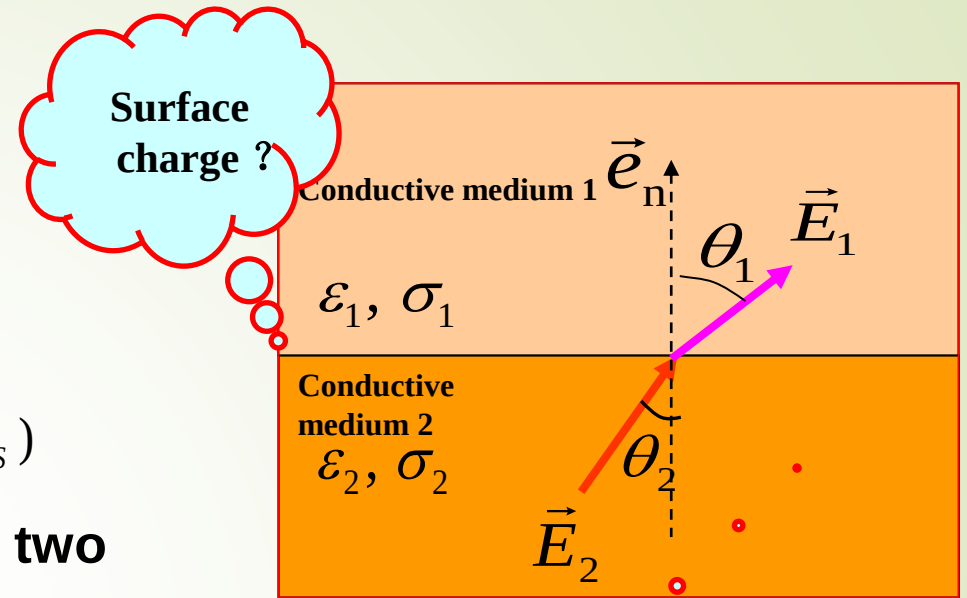
- Direction relation of the field vector at the two sides of the interface

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{E_{1t} / E_{1n}}{E_{2t} / E_{2n}} = \frac{\sigma_1 / J_{1n}}{\sigma_2 / J_{2n}} = \frac{\sigma_1}{\sigma_2}$$

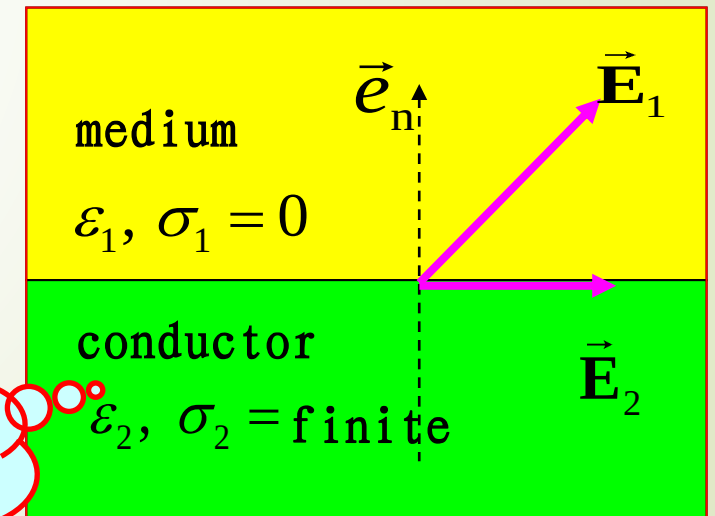
■ Medium and conductor

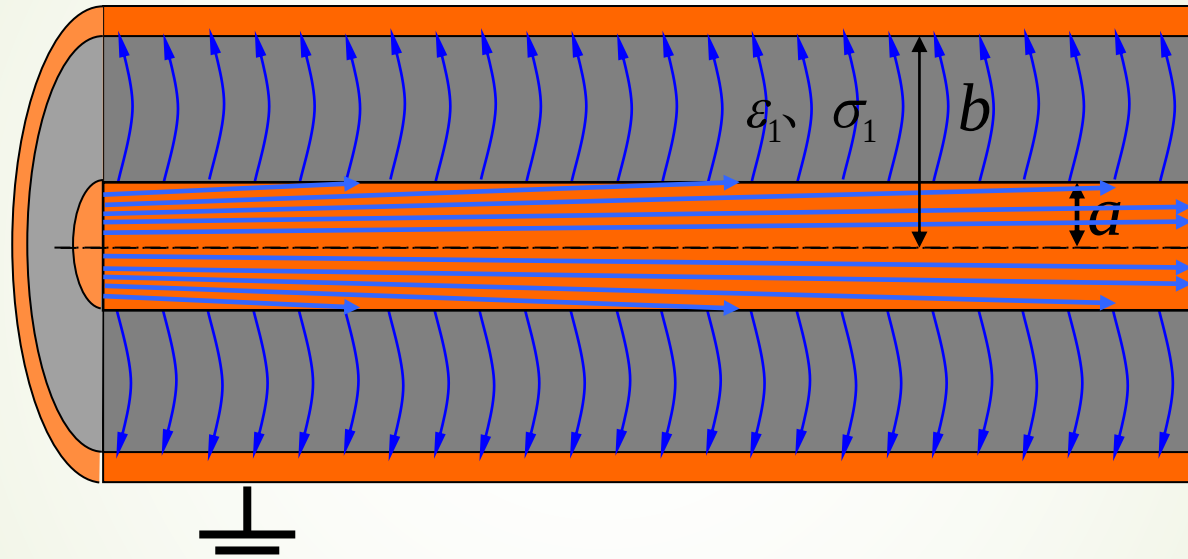
- The particularity of the field vector on both sides of the interface

$$\begin{cases} D_{1n} = \rho_s \\ E_{2n} = 0 \end{cases}$$



$$\sigma_2 \gg \sigma_1?$$





When will the steady current field be generated?

There is an electric field in a conducting medium !

The key is to find the electric field intensity

- **Based on the known charge**
- **Based on current (Ohm's law) ✓**
- **Based on electric potential ✓**

$$\nabla^2 \phi = 0$$

■ The method based on current to solve the problems of steady electric field

(1) The electric field intensity is expressed by the Ohm's Law (constitutive relation of the conducting medium).

$$\vec{E} = \frac{\vec{J}}{\sigma}$$

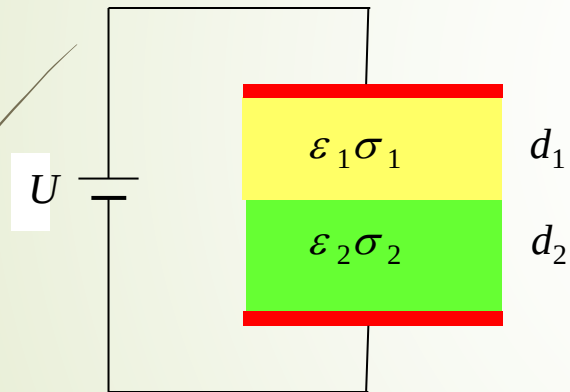
(2) The unknown quantity is expressed by a known quantity (usually the excitation voltage)

$$U = \int_P^Q \vec{E} \cdot d\vec{l}$$



Example

Ex. A parallel plate capacitor consists of two imperfect dielectrics in series. Their permittivity are ε_1 and ε_2 , the conductivities are σ_1 and σ_2 , and the thickness are d_1 and d_2 , respectively. If the impressed voltage is U , find the electric field intensities, the electric energies per unit volume, and the power dissipations per unit volume in two dielectrics.



Solution: Since no current exists outside the capacitor, the electric current lines in the capacitor can be considered to be perpendicular to the boundaries. Then we have

$$E_1 \sigma_1 = E_2 \sigma_2$$

$$E_1 d_1 + E_2 d_2 = U$$

In view of this we find

$$E_1 = \frac{\sigma_2}{d_1 \sigma_2 + d_2 \sigma_1} U$$

$$E_2 = \frac{\sigma_1}{d_1 \sigma_2 + d_2 \sigma_1} U$$

$$\rho_S = D_{1n} - D_{2n} = \frac{\sigma_2 \varepsilon_1 - \sigma_1 \varepsilon_2}{\sigma_2 d_1 + \sigma_1 d_2} U$$

Example

The **electric energies** per unit volume in two dielectrics, respectively, are

$$w_{e1} = \frac{1}{2} \varepsilon_1 E_1^2, \quad w_{e2} = \frac{1}{2} \varepsilon_2 E_2^2$$

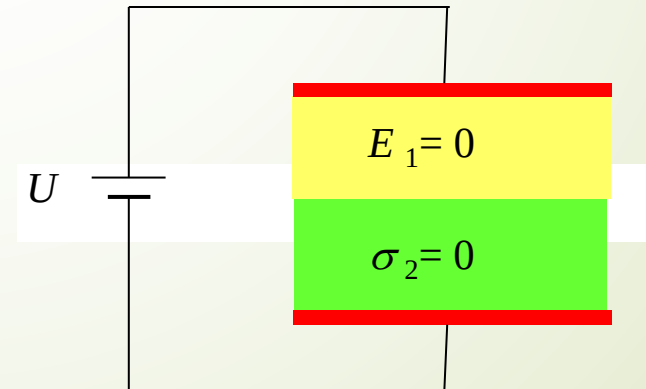
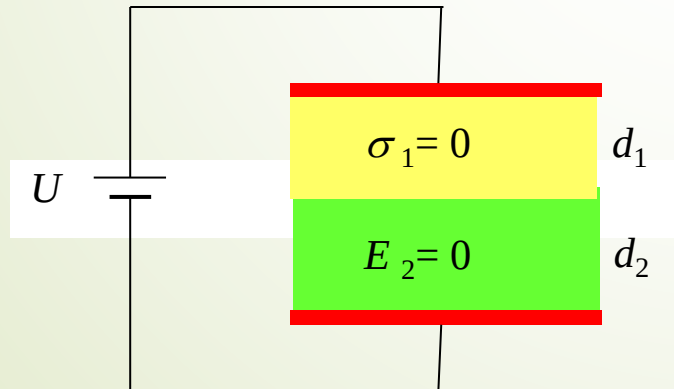
The **power dissipations** per unit volume in two dielectrics, respectively, are

$$p_{l1} = \sigma_1 E_1^2, \quad p_{l2} = \sigma_2 E_2^2$$

Two special cases are worth noting:

If $\sigma_1 = 0$, then $E_1 = \frac{U}{d_1}$, $E_2 = 0$, $w_{e2} = 0$, $p_{l2} = 0$.

If $\sigma_2 = 0$, then $E_1 = 0$, $w_{e1} = 0$, $p_{l1} = 0$, $E_2 = \frac{U}{d_2}$.

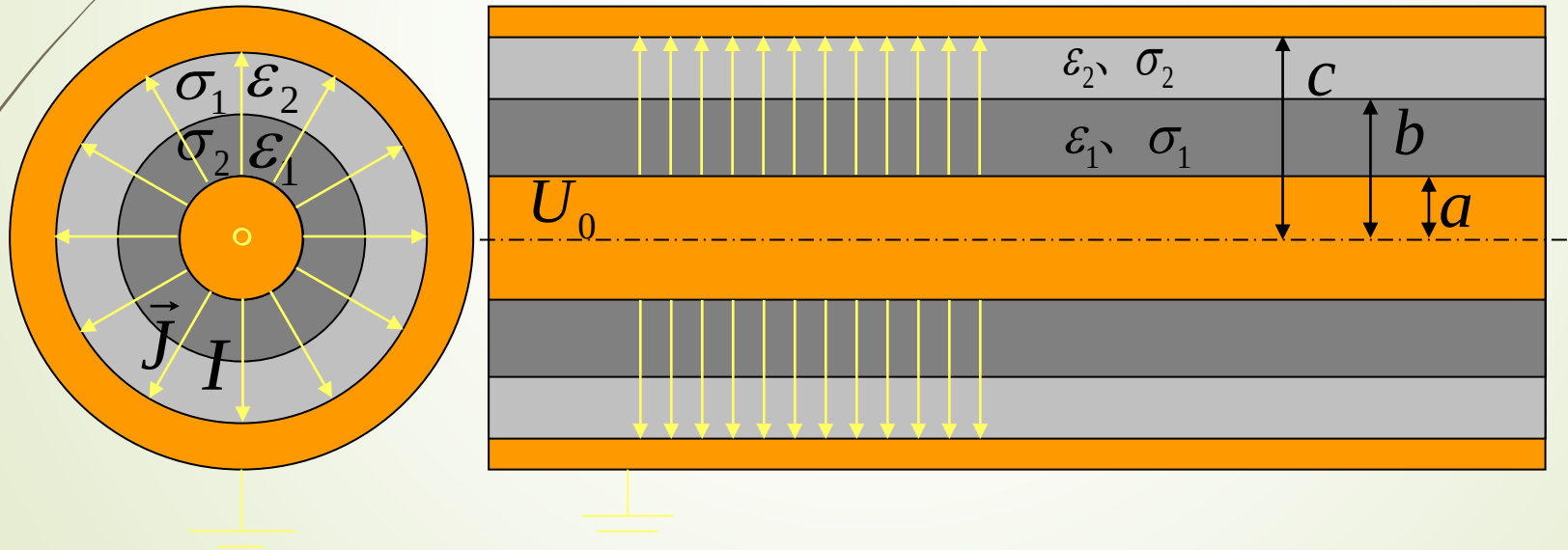


Example

Ex. The voltage of inner conductor is U_0 , outer conductor is grounded.

Find:

- (1) The current density and electric field in each region of coaxial line;
- (2) The surface density of the free charge on the interface.



Example

Solution: The current is flowing from the inner conductor to the outer conductor, only the normal component exists at the interface, so the current density is axial symmetrical distributed.

(1) The radial current in the coaxial cable is I ,

from $\int_S \vec{J} \cdot d\vec{S} = I$, the current density is

$$\vec{J} = \vec{e}_\rho \frac{I_l}{2\pi\rho} \quad (a < \rho < c)$$

Current per unit length

Electric field in medium 1

$$\vec{E}_1 = \frac{\vec{J}}{\sigma_1} = \vec{e}_\rho \frac{I_l}{2\pi\sigma_1\rho} \quad (a < \rho < b)$$

$$\vec{E}_2 = \frac{\vec{J}}{\sigma_2} = \vec{e}_\rho \frac{I_l}{2\pi\sigma_2\rho} \quad (b < \rho < c)$$



Example

Since
$$U_0 = \int_a^b \vec{E}_1 \cdot d\vec{\rho} + \int_b^c \vec{E}_2 \cdot d\vec{\rho} = \frac{I_l}{2\pi\sigma_1} \ln\left(\frac{b}{a}\right) + \frac{I_l}{2\pi\sigma_2} \ln\left(\frac{c}{b}\right)$$

So
$$I_l = \frac{2\pi\sigma_1\sigma_2 U_0}{\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)}$$

The current density and electric field inside two mediums:

$$\vec{J} = \vec{e}_\rho \frac{\sigma_1\sigma_2 U_0}{\rho[\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)]} \quad (a < \rho < c)$$

$$\vec{E}_1 = \vec{e}_\rho \frac{\sigma_2 U_0}{\rho[\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)]} \quad (a < \rho < b)$$

$$\vec{E}_2 = \vec{e}_\rho \frac{\sigma_1 U_0}{\rho[\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)]} \quad (b < \rho < c)$$



Example

The surface charge density on the inner surface of medium 1

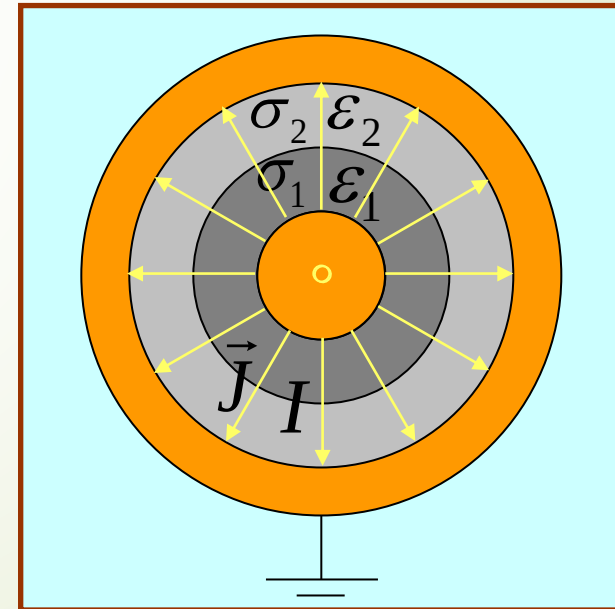
$$\rho_{s1} = \varepsilon_1 \vec{e}_\rho \cdot \vec{E}_1 \Big|_{\rho=a} = \frac{\varepsilon_1 \sigma_2 U_0}{a[\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)]}$$

The surface charge density on the outer surface of medium 2

$$\rho_{s2} = -\varepsilon_2 \vec{e}_\rho \cdot \vec{E}_2 \Big|_{\rho=c} = -\frac{\varepsilon_2 \sigma_1 U_0}{c[\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)]}$$

The surface charge density between two conducting mediums

$$\begin{aligned} \rho_{s12} &= (\varepsilon_2 \vec{e}_\rho \cdot \vec{E}_2 - \varepsilon_1 \vec{e}_\rho \cdot \vec{E}_1) \Big|_{\rho=b} \\ &= \frac{(\varepsilon_2 \sigma_1 - \varepsilon_1 \sigma_2) U_0}{b[\sigma_2 \ln(b/a) + \sigma_1 \ln(c/b)]} \end{aligned}$$



5.7 Resistance Calculations

► Resistance and conductance

$$R = \frac{1}{G} = \frac{U}{I}$$

$$G = \frac{I}{U}$$

❖ Method one of calculating conductance

1. Assuming the current between the two electrodes is I ;
2. For $U = \int_1^2 \vec{E} \cdot d\vec{l}$, calculate the potential difference between two conductors
3. According to the definition of the conductance to calculate the conductance : $G = I / U$



❖ Method two of calculating conductance

1. Assuming the potential difference between the two electrodes is U ;
2. For $I = \int_S \vec{J} \cdot d\vec{S}$, find the current between two conductors
3. According to the definition of the conductance to calculate the conductance : $G = I / U$

❖ Method three of calculating conductance

Electrostatic
simulation :

$$\frac{G}{C} = \frac{\sigma}{\varepsilon}$$



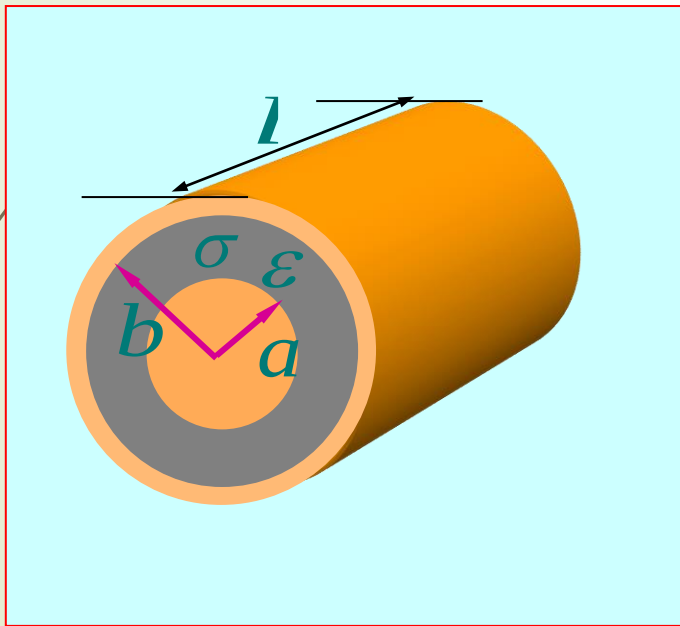
Comparison between steady electric field and electrostatic field with electrostatic simulation method

	electrostatic field $\rho = 0$ (area)	Steady electric field (outside the power supply)					
Basic equations	$\oint_S \vec{D} \cdot d\vec{S} = 0, \quad \oint_C \vec{E} \cdot d\vec{l} = 0$	$\oint_S \vec{J} \cdot d\vec{S} = 0, \quad \oint_C \vec{E} \cdot d\vec{l} = 0$					
	$\nabla \cdot \vec{D} = 0, \quad \nabla \times \vec{E} = 0$	$\nabla \cdot \vec{J} = 0, \quad \nabla \times \vec{E} = 0$					
Constitutive relations	$\vec{D} = \epsilon \vec{E}$	$\vec{J} = \sigma \vec{E}$					
Potential functions	$\vec{E} = -\nabla \varphi, \quad \nabla^2 \varphi = 0$	$\vec{E} = -\nabla \varphi, \quad \nabla^2 \varphi = 0$					
boundary conditions	$E_{1t} = E_{2t} \quad D_{1n} = D_{2n}$	$E_{1t} = E_{2t} \quad J_{1n} = J_{2n}$					
	$\varphi_1 = \varphi_2, \quad \epsilon_1 \frac{\partial \varphi_1}{\partial n} = \epsilon_2 \frac{\partial \varphi_2}{\partial n}$	$\varphi_1 = \varphi_2, \quad \sigma_1 \frac{\partial \varphi_1}{\partial n} = \sigma_2 \frac{\partial \varphi_2}{\partial n}$					
Corresponding physical quantity	electrostatic field	$\vec{E}$	$\vec{D}$	φ	Q	ϵ	C
	steady electric field	$\vec{E}$	$\vec{J}$	φ	I	σ	G

Example

Ex. The inner and outer radius is a 、 b , length is l , the medium between has permittivity of ϵ and conductivity of σ . Find the resistance of coaxial line

Solution: Assume the current flowing from the inner conductor to the outer conductor is I .



$$I \Rightarrow J = \frac{I}{2\pi\rho l} \Rightarrow E = \frac{J}{\sigma} = \frac{I}{2\pi\rho l\sigma}$$

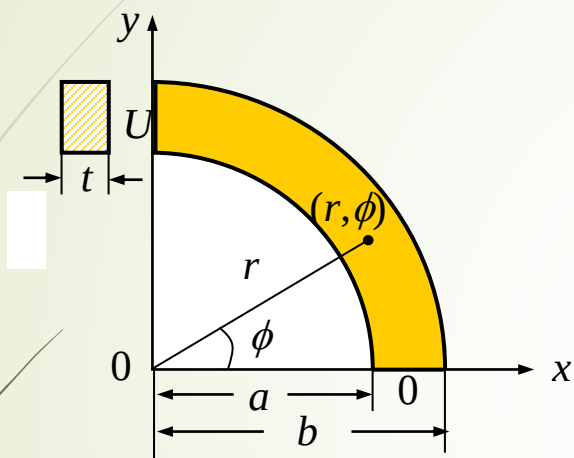
$$U = \int \vec{E} \cdot d\vec{l} = \int_a^b \frac{I}{2\pi\rho l\sigma} d\rho = \frac{I}{2\pi\sigma l} \ln \frac{b}{a}$$

$$G = \frac{I}{U} = \frac{2\pi\sigma l}{\ln(b/a)}$$

Resistance $R = \frac{1}{G} = \frac{1}{2\pi\sigma l} \ln \frac{b}{a}$

Ex. A quarter of a flat circular conducting washer is shown in the figure. Calculate the resistance between two end faces.

Example



Solution: The cylindrical coordinate system should be selected. Assume the electric potential difference between two end faces is U , and let

The electric potential $\varphi_1 = 0$ at $\phi = 0$

The electric potential $\varphi_2 = U$ at $\phi = \frac{\pi}{2}$

Since the electric potential φ is related to the angle ϕ , it should satisfy the following equation

$$\frac{d^2 \varphi}{d\phi^2} = 0$$

Cylindrical Coordinates:

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

The general solution is

$$\varphi = C_1 \phi + C_2$$

Example

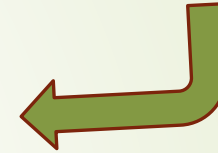
Based on the given boundary conditions, we find

$$\varphi = \frac{2U}{\pi} \phi$$

The current density $\mathbf{J}$ in the conducting medium is

$$\mathbf{J} = \sigma \mathbf{E} = -\sigma \nabla \varphi = -\mathbf{e}_\phi \sigma \frac{\partial \varphi}{r \partial \phi} = -\mathbf{e}_\phi \frac{2\sigma U}{\pi r}$$

$$\nabla u = \vec{e}_\rho \frac{\partial u}{\partial \rho} + \vec{e}_\phi \frac{1}{\rho} \frac{\partial u}{\partial \phi} + \vec{e}_z \frac{\partial u}{\partial z}$$



Then the current / flowing into the conducting medium across the end face at $\phi = \frac{\pi}{2}$ is

$$I = \int_S \mathbf{J} \cdot d\mathbf{S} = \int_S \left(-\mathbf{e}_\phi \frac{2\sigma U}{\pi r} \right) \cdot (-\mathbf{e}_\phi t dr) = \frac{2\sigma U t}{\pi} \int_a^b \frac{dr}{r} = \frac{2\sigma U t}{\pi} \ln \left(\frac{b}{a} \right)$$

Consequently, the resistance R between two end faces is

$$R = \frac{U}{I} = \frac{\pi}{2\sigma t \ln \left(\frac{b}{a} \right)}$$



Example

Ex. Annular conducting medium with thickness of h and two radius of r_1 and r_2 , has angle of ϕ_0 . The conductivity of the conductive medium is σ . find the resistance between the two electrodes along the ϕ direction.

Solution: The voltage between two electrodes along ϕ direction is U_0 , then current is flowing along ϕ direction, and current density is changed with ρ . It is easy to determine that the potential ϕ is just a function variables ϕ , thus the potential function satisfies the one-dimensional

Laplace equation
$$\nabla^2 \phi = \frac{1}{\rho^2} \frac{\partial^2 \phi}{\partial \phi^2} = 0$$

The general solution:

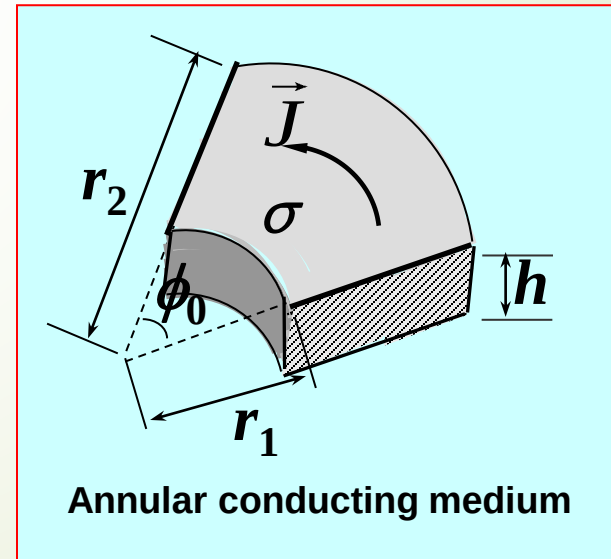
$$\phi = C_1 \phi + C_2$$

use B.C

$$\phi \Big|_{\phi=0} = U_0, \quad \phi \Big|_{\phi=\phi_0} = 0$$

then

$$C_1 = -U_0 / \phi_0, \quad C_2 = U_0$$



Example

So

$$\varphi = U_0 - \frac{U_0}{\phi_0} \phi$$

$$\vec{E} = -\nabla \varphi = -\vec{e}_\phi \frac{\partial \varphi}{\rho \partial \phi} = \vec{e}_\phi \frac{U_0}{\rho \phi_0}$$

Current density

$$\vec{J} = \sigma \vec{E} = -\vec{e}_\phi \frac{\sigma U_0}{\rho \phi_0}$$

The current between two plates

$$I = \int_S \vec{J} \cdot d\vec{S} = \int_{r_1}^{r_2} \vec{e}_\phi \frac{\sigma U_0}{\rho \phi_0} \cdot \vec{e}_\phi h d\rho = \frac{\sigma U_0 h}{\phi_0} \ln \frac{r_2}{r_1}$$

So the resistance between two plates along ϕ direction is:

$$R = \frac{U_0}{I} = \frac{\phi_0}{\sigma h \ln(r_2 / r_1)} \quad (\Omega)$$





Typical phenomena and conclusions of static electric field

—、Field is generated by static charge (electrostatic field)

- The electric field is zero inside the conductor($\sigma \neq 0$)
- The tangential electric field is zero on the surface of a conductor
 $\Rightarrow$ **equal potential body**
- The charge is zero inside the conductor
- Charge can only be located on the surface of the conductor, and it is concentrated in the sharp part of the surface.
- Application: electrostatic induction, electrostatic shielding, lightning rod ,



Typical electrostatic field

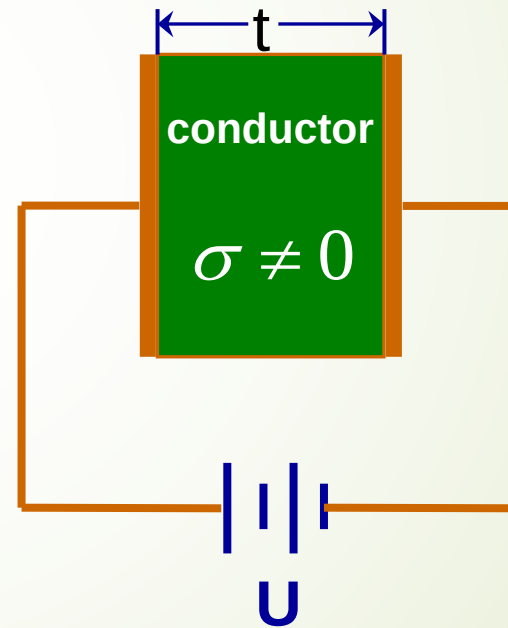
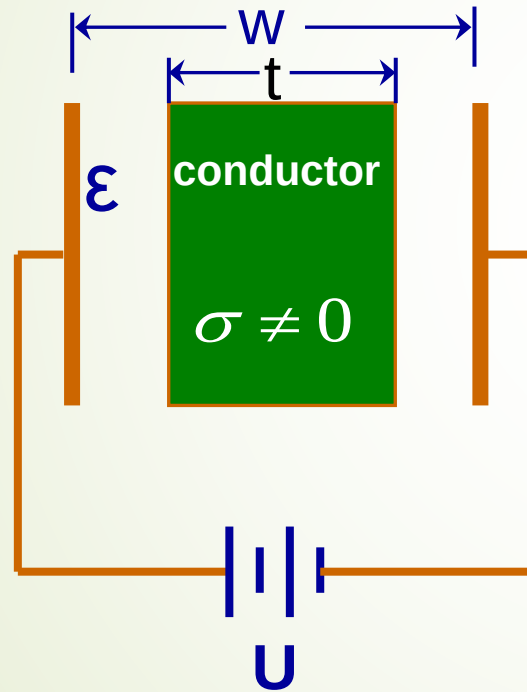
二、The DC electric field of the moving charge (steady electric field)

- There may be electric field $\vec{E} = \frac{\vec{J}}{\sigma}$ inside the conductor ($\sigma \neq 0, \infty$)
- The tangential electric field is not zero on the surface of the conductor.
 $\Rightarrow$ **Non equal potential body**
- There may be moving charge inside the conductor, but the net charge is zero.
- Net charge can only be located on the surface of the conductor
- The electric field and current inside the perfect conductor ($\sigma = \infty$), is zero .
- Electric field on the boundary of ideal conductor is Perpendicular to the surface
 $\Rightarrow$ **equal potential body**



Further understanding of the electrostatic field and steady electric field

QUESTIONS :



Find: 1) $\vec{E}, \varphi = ?$ 2) Energy storage or power consumption?

Summary

1. Basic equations (general problems)

$$\begin{cases} \nabla \cdot \vec{J} = 0 \\ \nabla \times \vec{E} = 0 \end{cases} \quad (\nabla \cdot \vec{D} = \rho) \quad \begin{matrix} \text{Constitutive relations:} \\ \vec{J} = \sigma \vec{E} \quad (\vec{D} = \varepsilon \vec{E}) \end{matrix}$$

2. Boundary conditions (general problems)

$$\begin{cases} J_{1n} - J_{2n} = 0 \\ E_{1t} - E_{2t} = 0 \end{cases} \quad D_{1n} - D_{2n} = \rho_s$$

3. Resistance and conductance

electrostatic
simulation method

$$\frac{G}{C} = \frac{\sigma}{\varepsilon}$$



Summary

Constitutive relations :

2. Boundary conditions (general problems)

1. Basic equations (general problems)





Homework

5-1, 5-6, 5-10, 5-15, 5-16,
5-22